



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Balaih
Student Name	V.Sathvik
Roll Number	2520090097
Branch	CSIT
Course Outcome	CO4

Case Study Topic: Google Maps Live Re-Routing using Dijkstra's Algorithm

Work Done Summary:

This case study demonstrates the use of Dijkstra's Algorithm for finding the shortest route between locations in a road network. The impact of a road closure was analyzed by recomputing shortest paths and comparing route changes.

Implementation Details:

- Created a weighted graph representing road locations.
- Added vertices and connecting roads.
- Assigned travel-time weights to each road.
- Selected KOR as the source location.
- Selected EC as the destination location.
- Initialized distances to infinity.
- Set source distance to zero.
- Used a Priority Queue as a min-heap.
- Extracted the minimum-distance vertex.
- Relaxed adjacent edges.
- Updated distances when shorter paths were found.
- Stored parent nodes for path recovery.
- Computed the shortest route from KOR to EC.
- Simulated road closure between MGR and HSR.

- Recomputed shortest paths and compared results.

Results / Output:

```
DIJKSTRA SHORTEST PATH  
  
Source : KOR  
Destination : EC  
  
Shortest Distance = 14  
  
Path:  
KOR -> JAY -> MGR -> EC  
  
Route Computation Successful.
```

Conclusion:

The case study demonstrated that Dijkstra's Algorithm efficiently computes shortest paths in road networks with non-negative edge weights. When a road closure occurs, shortest paths can be recomputed to provide updated routes. Therefore, Dijkstra's Algorithm is suitable for navigation and live route-planning systems.

GitHub Repository Link:

https://github.com/vemulapallisathvik-lgtm/DSA-2_CASESTUDY

Student Signature	Faculty Signature